

resistor is connected to GP26 pin of Board1 for interfacing external sensors.

Software

The project is controlled by the Python program developed on Mu

editor. Mu editor is a simple code editor that works with Adafruit CircuitPython boards, including Raspberry Pi Pico, and on most operating systems, including Windows and Linux. CircuitPython is based on Python programming language

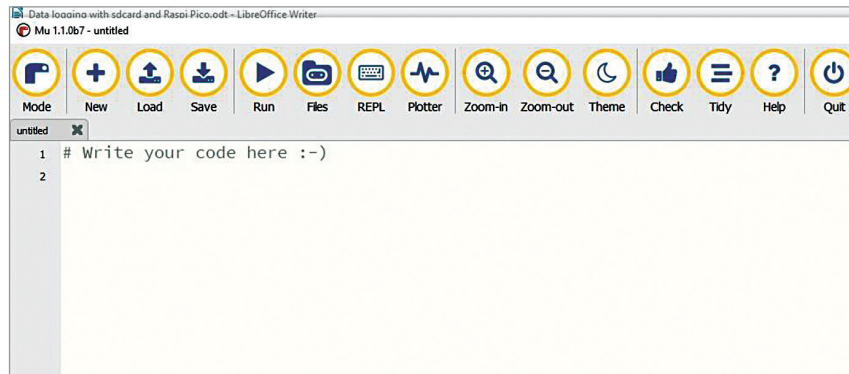


Fig. 3: Mu editor window

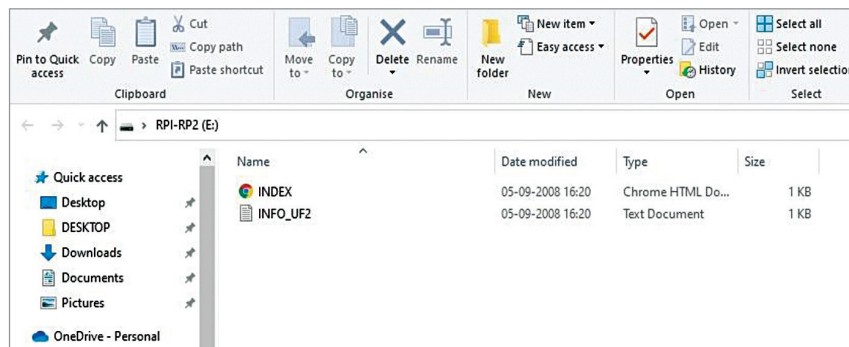


Fig. 4: New drive window

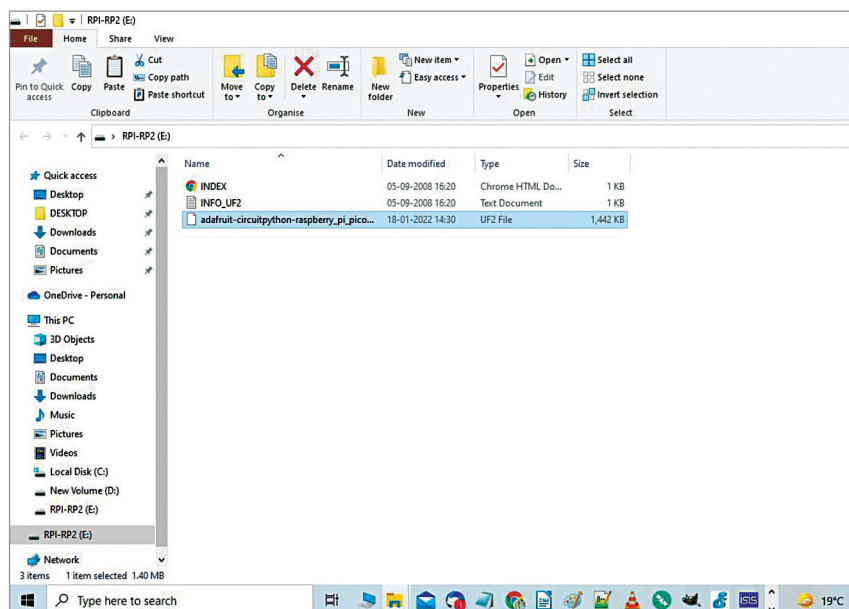


Fig. 5: CircuitPython file in the drive

and is easy to understand.

Both Mu and CircuitPython are required for this project. Mu editor can be downloaded from <https://codewith.mu> and then installed using instructions given at <https://learn.adafruit.com/getting-started-with-raspberry-pi-pico-circuitpython/installing-mu-editor>. After installing Mu editor, run it and you will see the window shown in Fig. 3.

CircuitPython version 7.1.1 was used during testing of this project. CircuitPython board requires a bootloader called UF2 (USB flashing format) that makes installing and updating CircuitPython a quick and easy process. Download the most recent version of CircuitPython bootloader .UF2 file from https://circuitpython.org/board/raspberry_pi_pico/

Installing CircuitPython on the board is easy. Just copy and paste the file as explained here. First press the BOOTSEL button on the Raspberry Pi Pico board, while inserting the USB cable of the RPI board to laptop/PC, and release it after a new drive (RPI-RP2) window appears, as shown in Fig. 4.

Next, copy *adafruit-circuitpython-raspberry_pi_pico-en_US-7.1.1.uf2* file downloaded in previous step. Paste it into the new window, as shown in Fig. 5. Now open Mu editor window and close all other windows.

Next, the first thing to do in Mu is to select 'Mode' from the main menu. In this project, you should select RP2040 option in Select Mode window, as shown in Fig. 6.

Start coding on the Mu editor window or load the existing code. You can load the *analog_sensor.py* code that comes along with this project. This code will work with either LDR or force sensor but not with both at the same time.

Connect LDR to the circuit, as shown in the schematic, and run the code by clicking Run button. When you cover the LDR with an opaque